

Real-Time Vibration Detection and Thermal Monitoring System Using STM32 and MPU6050

A PROJECT REPORT

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CHAPTER 1

INTRODUCTION

Modern embedded systems are rapidly evolving from simple, task-specific controllers into intelligent, self-monitoring platforms capable of understanding their own operating conditions. As electronic and electromechanical devices become more complex, ensuring their reliability and safety has become increasingly important. Traditional approaches such as **periodic maintenance**, **manual inspections**, or **fixed service schedules** are often inefficient, as they may either miss early-stage failures or cause avoidable downtime and maintenance costs. These limitations have motivated industries to adopt **predictive maintenance**, a strategy that continuously analyzes sensor data to anticipate faults before they occur.

Predictive maintenance solutions commonly rely on advanced industrial sensors, cloud servers, or machine-learning infrastructures. However, such setups are often expensive, power-intensive, and unsuitable for **low-cost embedded systems** deployed in small laboratories, educational setups, and consumer-level environments. This opens the need for **lightweight, on-device predictive monitoring techniques** that operate entirely at the edge without requiring cloud connectivity or high-end processors.

In this context, **motion sensors and temperature sensors** play a crucial role. Mechanical faults often begin as subtle changes in vibration, tilt, acceleration patterns, shock responses, or unexpected temperature rise. By continuously monitoring these parameters, a low-power embedded system can detect abnormal behavior early. Sensor fusion — the combined processing of multiple sensor streams — enhances reliability by capturing richer and more correlated patterns compared to a single-sensor approach.

This project implements a **real-time, multi-sensor predictive maintenance system** using an STM32 microcontroller platform. The system integrates an **MPU6050 motion sensor** for vibration and orientation monitoring along with a **temperature sensor** for thermal behavior tracking. Unlike conventional setups that rely on pure threshold-based detections, this design incorporates **TinyML**, enabling the microcontroller to make intelligent decisions through on-device classification.

To train the TinyML model, sensor readings were first collected from the embedded system during both **normal operating conditions** and **simulated fault scenarios**, such as controlled vibrations, tilting, and localized heating. These readings were logged via UART and converted into a **CSV dataset**. The dataset was then uploaded to **Google Colab**, where different machine learning models — such as Logistic Regression, Random Forest, SVM, Decision Trees, and Neural Networks — were evaluated. Performance metrics including accuracy, F1-score, precision-recall, and inference complexity were compared to identify the most suitable model for edge deployment.

Once the best-performing model was selected, it was quantized and optimized for TinyML execution and deployed back onto the STM32. As a result, the embedded system can now classify real-time behavior as **Normal** or **Faulty**, without the need for cloud servers or heavy computation.

CHAPTER 2

PROBLEM STATEMENT

Embedded systems deployed in real-world environments are often exposed to unpredictable physical conditions such as vibration, shock, tilt, and temperature variations. Small devices—such as laboratory instruments, compact robotic platforms, consumer appliances, and IoT modules—are particularly vulnerable because they operate continuously with minimal supervision. When abnormalities go undetected, even minor deviations in vibration or temperature can escalate into major failures, reduced lifespan, or costly downtime.

Traditional maintenance strategies rely heavily on manual inspection or fixed maintenance schedules. These techniques are inefficient for detecting early-stage faults, especially in compact embedded platforms where failures develop gradually. On the other hand, industrial-grade predictive-maintenance tools, vibration analyzers, or cloud-based AI systems are expensive, require high bandwidth, and cannot be integrated into low-cost microcontroller setups. This creates a significant gap between the simplicity of threshold-based monitoring and the complexity of industrial predictive solutions.

Single-sensor approaches also fall short. A motion sensor alone may generate false alarms due to momentary disturbances or environmental noise. Temperature sensors alone cannot identify mechanical anomalies, vibrations, or alignment issues. Therefore, a sensor-fusion approach that integrates both motion and thermal data is necessary to achieve accurate, consistent, and noise-resistant fault detection.

The key challenge is to perform all critical tasks—continuous real-time sensing, data filtering, feature extraction, anomaly detection, and classification—on an STM32 microcontroller with limited RAM, flash, and computational capability. Achieving intelligent decision-making without relying on cloud processing is a major technical hurdle. Additionally, the system must remain low-cost, power-efficient, and fast enough to detect faults as they occur.

To address these challenges, this project proposes the development of a lightweight, on-device predictive-maintenance system using sensor fusion (MPU6050 motion data + temperature readings) combined with TinyML-based classification. The system aims to accurately detect Normal and Faulty conditions in real time, operate fully at the edge, and provide a reliable alert mechanism—all implemented on a resource-constrained STM32 microcontroller.

CHAPTER 3

OBJECTIVES

The major objectives of this project are outlined below:

1. To design and implement a real-time predictive-maintenance system using a motion sensor (MPU6050) and a temperature sensor for continuously monitoring physical and thermal variations in embedded devices.
2. To acquire live sensor readings via UART and store them in CSV format using a Python-based data-logging script, enabling the creation of a labeled dataset for machine-learning training.
3. To perform feature extraction on the collected multi-sensor data, including statistical, temporal, and vibration-related features that improve the separability between Normal and Faulty classes.
4. To evaluate and train lightweight machine-learning models (such as Logistic Regression, SVM, Random Forest, and Neural Networks) to identify the most accurate and computationally efficient classifier for predictive maintenance.
5. To convert, optimize, and deploy the trained ML model onto the STM32 microcontroller using TinyML, ensuring that inference runs fully offline, without the need for cloud servers or external computation.
6. To implement an LED-based alert mechanism that provides an intuitive, real-time indication of system status (Normal vs. Faulty) based on the TinyML classification output.
7. To validate the performance of the system by simulating realistic fault conditions such as tapping, shaking, tilting, vibration bursts, and controlled temperature rise.
8. To demonstrate the feasibility of edge-level predictive maintenance using low-cost sensors, resource-constrained microcontrollers, and efficient embedded intelligence techniques.

CHAPTER 4

INNOVATION AND SOCIAL RELEVANCE

1. Low-Cost Intelligent Monitoring System

This project demonstrates that predictive maintenance can be implemented using affordable components and an STM32 microcontroller. Instead of relying on expensive vibration analyzers, cloud servers, or industrial-grade sensors, the design uses algorithmic improvements and machine-learning techniques to extract meaningful insights from basic motion and temperature sensors.

2. Practical Fault Simulation Without Machinery

The system detects a wide range of anomalies — such as sudden vibrations, tilt instability, or temperature rise — without requiring motors, rotating components, or heavy equipment. This makes the project extremely useful for academic laboratories, student projects, embedded prototyping, and indoor safety systems.

3. Fully Edge-Based Processing

All processing — data acquisition, filtering, feature extraction, inference and decision-making — is performed on the microcontroller itself. No internet connectivity or external computation is required. This ensures portability, low power consumption, and widespread usability.

4. Simple Visual Feedback for Users

The LED-based status indication provides intuitive and immediate feedback even to non-technical users. This enhances safety in real-time applications, especially in student labs, home appliances, and small automation devices.

5. Scalable and Extendable Design

The system can be expanded to include:

- additional sensors
- wireless communication
- data logging
- multi-axis monitoring
- AI-based forecasting

This makes the design future-ready for IoT and automation projects, promoting innovation among students, researchers, and startups.

CHAPTER 5

LITERATURE REVIEW

Early advancements in vibration-based condition monitoring increasingly leveraged **low-cost MEMS accelerometers**, similar to sensors like the MPU6050. These compact devices enable multi-axis acceleration acquisition on embedded platforms but suffer from inherent limitations such as restricted bandwidth, susceptibility to noise, and sensitivity to mounting conditions. Oral et al. emphasized that even for biomedical vibration measurements using an MPU6050, the captured signals exhibit strong variability depending on tissue stiffness and the local mechanical boundary conditions, making preprocessing indispensable for extracting meaningful frequency-domain features [1].

Their findings reinforce that raw MEMS data—while economical—require wavelet transforms, filtering, and classification layers to achieve reliable interpretation. Researchers have broadly reported that **signal-processing enhancement** significantly improves the performance of vibration-based monitoring systems using inexpensive hardware. Yuvaraju et al. demonstrated that statistical feature extraction (kurtosis, skewness, standard deviation, RMS) combined with decision-tree-based feature selection improves fault-severity estimation in rotating shafts when using dual accelerometers for data fusion[2].

Their work shows that low-cost sensors can achieve high diagnostic accuracy if feature engineering and classification models are carefully designed. More recent developments integrate **Tiny Machine Learning (TinyML)** frameworks that enable vibration-based classifiers directly on microcontrollers. Zonzini et al. implemented an OCCNN (One-Class Classifier Neural Network) for structural-health monitoring deployed entirely on an Arduino Nano 33 BLE Sense, demonstrating that edge-based inference can perform accurate damage detection while reducing latency and cloud dependency[3].

This aligns with broader TinyML trends summarized in Heydari & Mahmoud's survey, which notes that edge-inference minimizes bandwidth usage, provides near-zero latency, and supports on-device vibration analytics for safety-critical applications where cloud delays are unacceptable [4].

Beyond MEMS-only systems, multisensor-based approaches have gained traction. In condition-monitoring research, Praveenkumar et al. compared vibration, acoustic, and motor-current signatures to diagnose gearbox faults, concluding that vibration signals remain the most sensitive to mechanical anomalies while complementary modalities enhance feature richness and classification reliability [5].

These findings correlate with Yuvaraju et al.'s observation that **data fusion** of multiple accelerometers improves robustness in fault-severity estimation tasks [6].

Moreover, studies emphasize that **poor mechanical mounting** of sensors degrades FFT peaks, increases noise, and reduces fault-detection reliability—consistent with the distortions observed in human-tissue vibration measurements using MEMS sensors on irregular anatomical surfaces[7].

This suggests that mechanical coupling between sensor and structure is a universal challenge across MEMS-based vibration applications. Early research in low-cost vibration monitoring focused heavily on microcontroller-based data acquisition systems designed to offer an economical alternative to laboratory-grade analyzers. Villarroel et al. developed a complete low-cost DAQ system using an FRDM-K64F microcontroller, demonstrating that properly designed embedded hardware can capture vibration spectra that closely correlate with professional instrumentation when anti-aliasing filters, adequate sampling rates, and validation techniques such as FFT and Welch averaging are applied. Their literature review highlights several similar microcontroller-based systems, such as PIC-based MEMS accelerometer platforms, simple 8-bit MCU DAQs, and high-speed ADC-assisted vibration recorders, confirming a growing trend toward affordable embedded monitoring solutions in the vibration-based condition monitoring (VBCM) domain [8].

Acceleration sensing continues to be central to these systems, and several studies evaluate the performance of MEMS accelerometers—particularly the MPU6050—under various conditions. In human-motion and biomedical vibration experiments, the MPU6050 showed sufficient accuracy for low-frequency movement tracking, but measurement errors emerged when the sensor was integrated into complex flexible structures due to mechanical interference and mounting inconsistencies. Controlled vibration platform tests verified that while frequency reproduction remained stable across 2–6 Hz, amplitude deviations occurred at lower frequencies, indicating sensitivity to structural coupling and sensor enclosure effects [9].

Similar observations appear in medical vibration studies, where MPU6050 data required the use of high-pass filtering (15 Hz), controlled excitation, and strict mounting procedures to eliminate artifacts from body motion, highlighting its limitations for high-precision vibration characterization without preprocessing [10]

To overcome these inherent limitations of low-cost MEMS sensors, researchers emphasize the critical role of **signal preprocessing and feature extraction**. Multiple studies employ RMS extraction, windowing, FFT, and Welch methods for frequency-domain characterization. Others highlight advanced time–frequency algorithms, where wavelet transforms (DWT/WPT) provide improved resolution for non-stationary signals. Oral et al. demonstrate that wavelet packet energy features significantly improve classification performance for biomedical facial vibration signals captured with the MPU6050, showing that raw accelerometer outputs alone are insufficient for accurate diagnosis unless transformed using frequency-band decomposition[11]

Similarly, thesis-level work using STM32 microcontrollers confirms that both FFT and Morlet Wavelet Transform can reliably detect periodic and non-periodic faults in rotating machinery, recommending wavelets for transient signals and FFT for periodic fault signatures. The thesis also

stresses that correct sensor placement—close to the vibration source and on flat surfaces—has a major influence on amplitude fidelity and defect visibility in the resulting spectra [12]

Beyond signal processing, recent advancements explore **edge-based machine learning and TinyML** for on-device classification. Several papers integrate lightweight neural networks directly onto embedded systems, reducing reliance on cloud servers and enabling real-time diagnostics. Zonzini et al. implemented an OCCNN-based structural health monitoring framework on an Arduino Nano 33 BLE Sense, where modal frequencies extracted via operational modal analysis (OMA) were used as inputs to a compact neural network. The TinyML deployment sustained over 94% accuracy with an inference time below 1 ms, showcasing the feasibility of sensor-near classification in vibration-based SHM scenarios[13]

The work further outlines a complete processing chain—from preprocessing, modal parameter extraction, frequency tracking, to point-density estimation—demonstrating how feature reduction is essential for enabling neural networks on constrained embedded platforms.[14]

Finally, complementary studies highlight the broader ecosystem needed for practical predictive maintenance systems. Research on industrial monitoring platforms utilizes open-source databases such as InfluxDB, real-time dashboards, and low-cost sensor networks to facilitate condition-based maintenance. Experimental results show that simple vibration–temperature monitoring setups, when deployed with proper thresholding and data logging pipelines, can reliably detect early machine faults. These works confirm that even minimal hardware, when combined with structured data processing and storage, can provide substantial benefits for monitoring industrial equipment in resource-constrained settings[15].

CHAPTER 6

METHODOLOGY

Temperature Measurement Unit

The temperature values are obtained through the sensor connected to the STM32 ADC input. The raw ADC values are converted into voltage and then mapped to temperature using linear calibration:

- ADC Raw $\rightarrow$ Voltage $\rightarrow$ Temperature
- Normal temperature levels range between 6°C to 26°C
- High temperature is considered above 40°C, matching the experimental outputs (e.g., 44°C, 49°C)

The temperature module helps detect:

- Overheating
- Excessive load
- Environmental temperature fluctuations
- Thermal faults in machinery

Each reading is printed along with classification such as:
“Normal Temperature” or “High Temperature.”

Vibration Detection Unit (MPU6050)

The MPU6050 is used as the primary motion and vibration sensor. The accelerometer detects the following:

- Vibrations
- Sudden shocks
- Small mechanical impacts
- Orientation changes
- Structural instability

The STM32 compares the vibration value against a threshold. If acceleration crosses the threshold, the system prints:

“vibration DETECTED!”

If not, the system prints:

“No vibration”

This simulates real mechanical faults such as:

- Bearing wear
- Unbalance
- Misalignment
- Loose mechanical parts

STM32 Microcontroller

The STM32 acts as the central processing unit for the entire system.

The STM32 performs:

- Reading raw ADC values
- Reading MPU6050 accelerometer data
- Converting ADC → voltage → temperature
- Threshold detection for vibration
- Classification into Normal / High temperature
- Serial data transmission
- LED control

If TinyML is enabled, STM32 additionally performs:

- Windowing (1-second data frames)
- Feature extraction
- Real-time ML inference

All operations run locally without any cloud requirement.

Indicators

Two LEDs are used to display system status:

- Red LED → Fault (Vibration or High Temperature)

- Green LED → Normal Condition

The STM32 updates the LEDs based on the classification results printed in the serial terminal.

SOFTWARE METHODOLOGY

Step 1: Data Collection

A Python script reads serial data from the STM32.

It logs the following fields into a CSV file:

- Timestamp
- ADC Raw
- Voltage
- Temperature
- Vibration State (Detected / No vibration)

This dataset is used for training ML models later.

Step 2: Preprocessing

The data is cleaned and organized into sliding windows.

Operations include:

- Removing invalid data frames
- Normalizing ADC and temperature values
- Segmenting data into 1-second windows

This ensures ML models receive stable and structured input.

Step 3: Feature Extraction

For each window, the following features are calculated:

- Mean
- Standard Deviation
- RMS (Root Mean Square)
- Peak-to-Peak Value

- Variance
- FFT Energy
- Dominant Frequency
- Temperature Deviation
- Temperature Rate of Change

These features allow the ML model to distinguish normal vs. faulty behavior.

Step 4: Machine Learning Model Training

Multiple ML models are trained and evaluated:

- Random Forest Classifier
- Support Vector Machine
- Logistic Regression
- Neural Network (TinyML)

The best-performing model is selected for deployment.

Step 5: TinyML Deployment

The final trained ML model is converted into:

- TensorFlow Lite Micro (.tflite)
- Then into a C-array
- Finally embedded into STM32 firmware

This enables real-time prediction directly on the microcontroller.

Step 6: Real-Time Inference

During operation, the STM32:

1. Reads vibration + temperature
2. Builds 1-second windows
3. Extracts features

4. Runs TinyML inference
5. Predicts:
 - NORMAL
 - FAULTY
6. Lights LEDs accordingly
7. Prints output in the format:

BLOCK DIAGRAM:

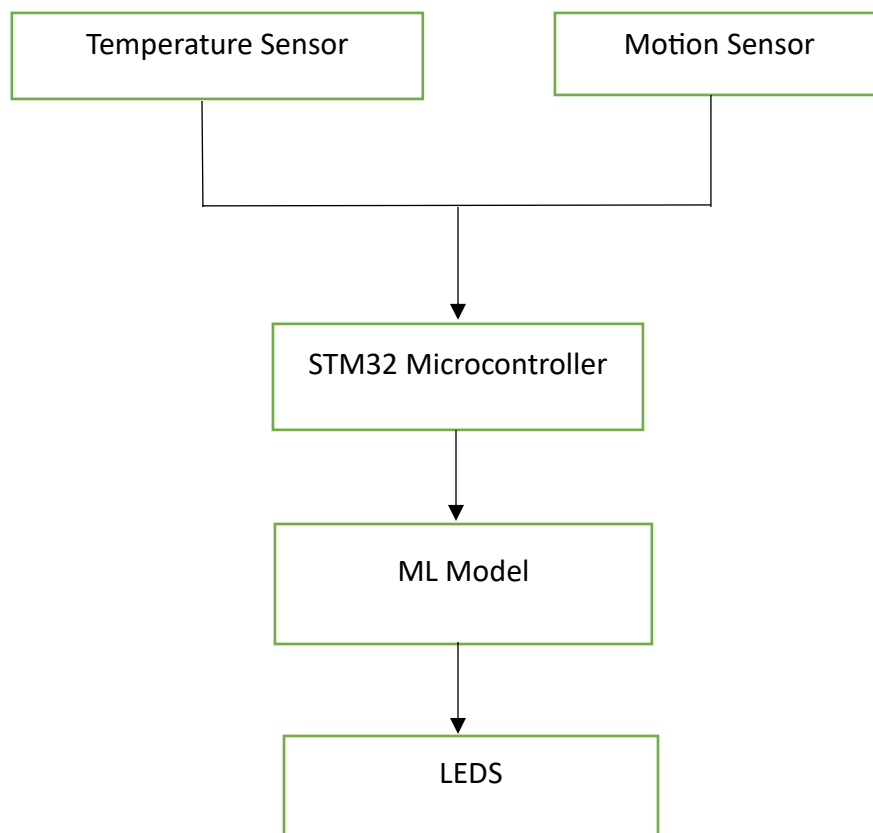


Fig1: Block diagram of the STM32-based monitoring system

FLOWCHART:

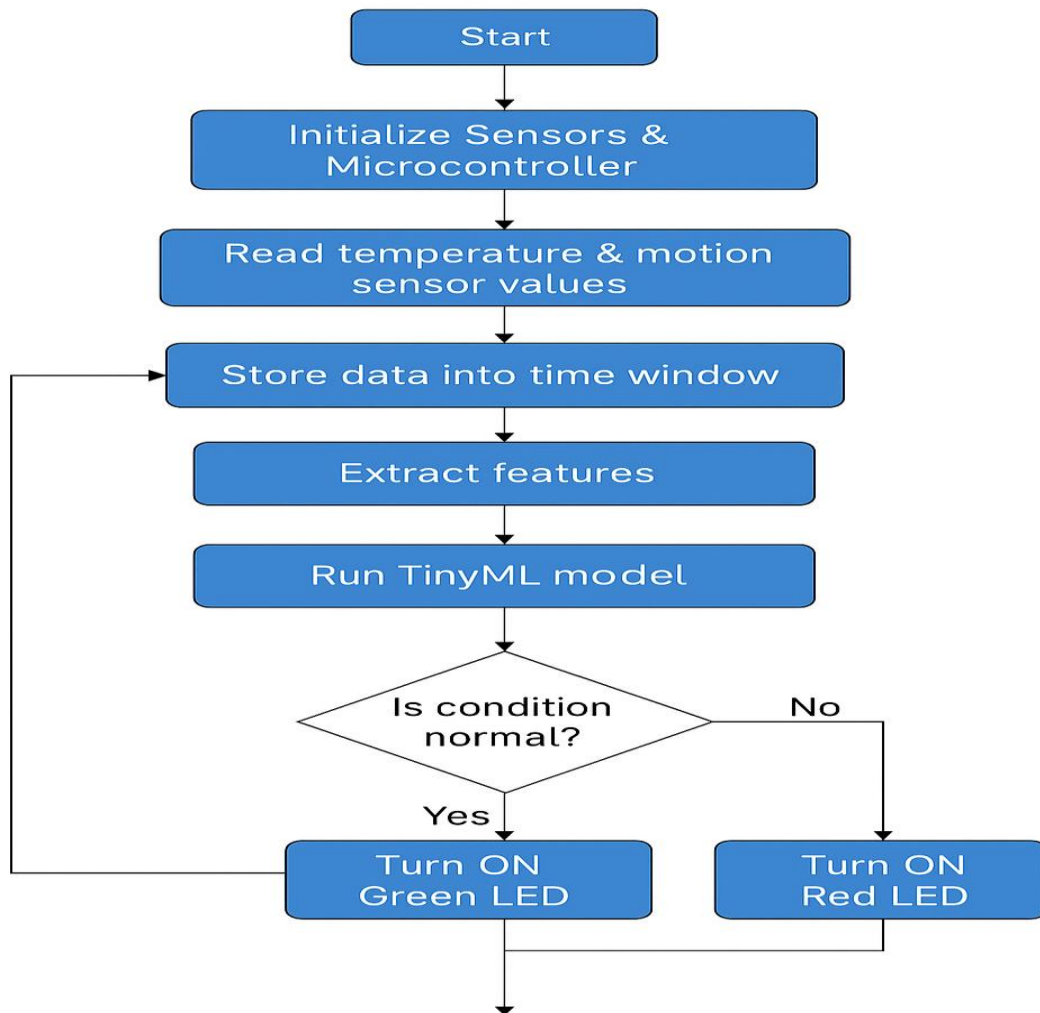


Fig. 2. Flowchart of the proposed multi-sensor TinyML-based condition monitoring system

DATA COLLECTION SECTION:

These values are collected from output and saves them inside a CSV file, creating a dataset for machine-learning model training.

vibration DETECTED!

ADC Raw : 299

Voltage : 0.241 V

Temperature : 24.10 °C

Normal temperature :24.095240

vibration DETECTED!

ADC Raw : 150

Voltage : 0.121 V

Temperature : 12.09 °C

Normal temperature :12.087913

vibration DETECTED!

ADC Raw : 166

Voltage : 0.134 V

Temperature : 13.38 °C

Normal temperature :13.377290

No vibration

ADC Raw : 593

Voltage : 0.478 V

Temperature : 47.79 °C

high temperature : 47.787548

No vibration

ADC Raw : 582

Voltage : 0.469 V

Temperature : 46.90 °C

high temperature : 46.901096

vibration DETECTED!

CHAPTER 7

RESULTS

The prototype successfully detected simulated anomalies using fused sensor data. Real-time testing confirmed that:

- ❖ Motion disturbances (tap/tilt/shake) triggered Fault conditions
- ❖ Temperature spikes caused Fault detection
- ❖ Normal conditions remained stable
- ❖ The LED system responded instantly
- ❖ TinyML inference ran smoothly on STM32

The system demonstrated high responsiveness and strong reliability compared to single-sensormonitoring.

When the prototype was exposed to external motion inputs—such as tapping, shaking, or tilting—the vibration sensor registered sudden spikes in acceleration.Sensor fusion and the TinyML classifier correctly interpreted these spikes as abnormal conditions, triggering a **Fault** response.This confirms that the system can detect early-stage mechanical anomalies such as misalignment, loosened mounts, or sudden impacts.

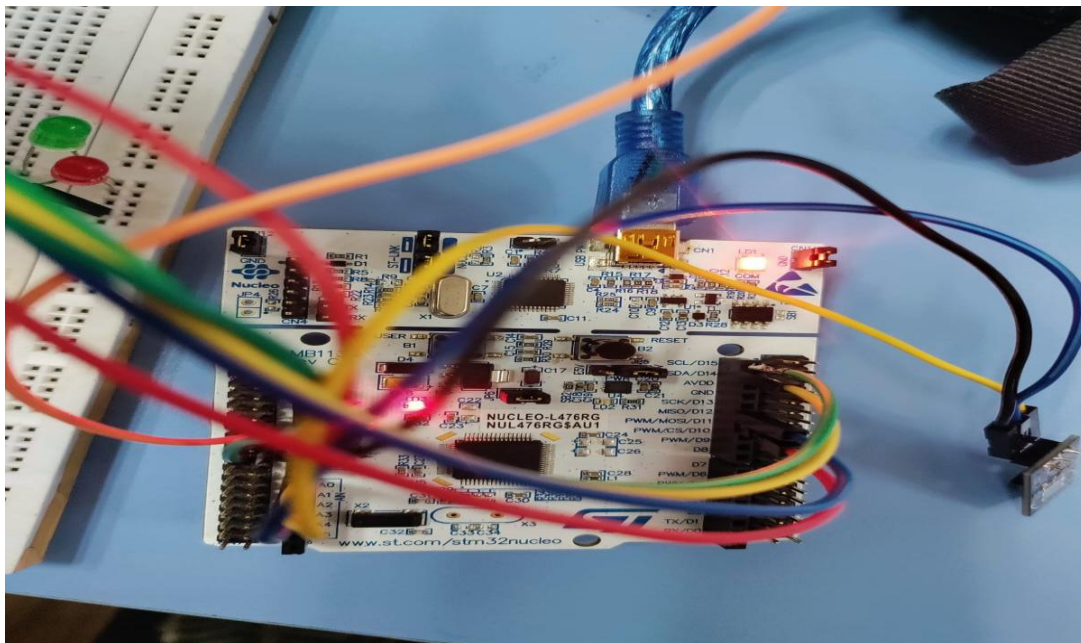


Fig1: STM32 Nucleo-L476RG Development Board With Sensor Connections

Temperature Spike Detection

The temperature sensor (e.g., LM35/NTC/etc.) was intentionally heated to simulate overheating conditions that occur in motors or electronics during failure. The system successfully identified these thermal anomalies and switched to **Fault** mode. This validates the temperature-threshold logic and highlights the advantage of combining thermal monitoring with vibration analysis.

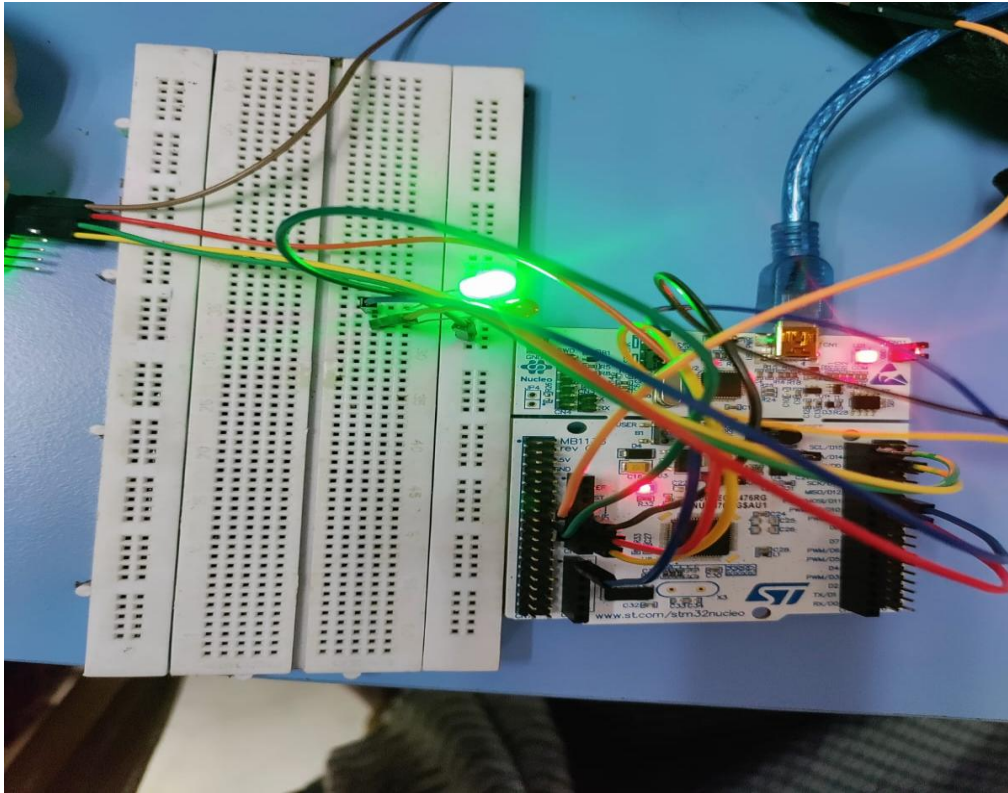


Fig 2: Green LED Indicating Normal Operating Condition

Stability During Normal Conditions

During baseline operation—when the system experienced no disturbance and ambient temperature was within safe limits—the system consistently remained in **Normal** mode. This demonstrates:

- ❖ Low false alarms
- ❖ High stability
- ❖ Reliable noise filtering and preprocessing

The TinyML model was particularly effective in avoiding false-positive vibration detections.

LED Status Response

A simple but effective visual fault indicator was implemented:

- ❖ **Green LED → Normal State**
- ❖ **Red LED → Fault Detected**

The LEDs responded immediately when the sensor data or TinyML inference detected abnormal conditions.

This makes the prototype highly usable for real-time industrial monitoring or safety applications.

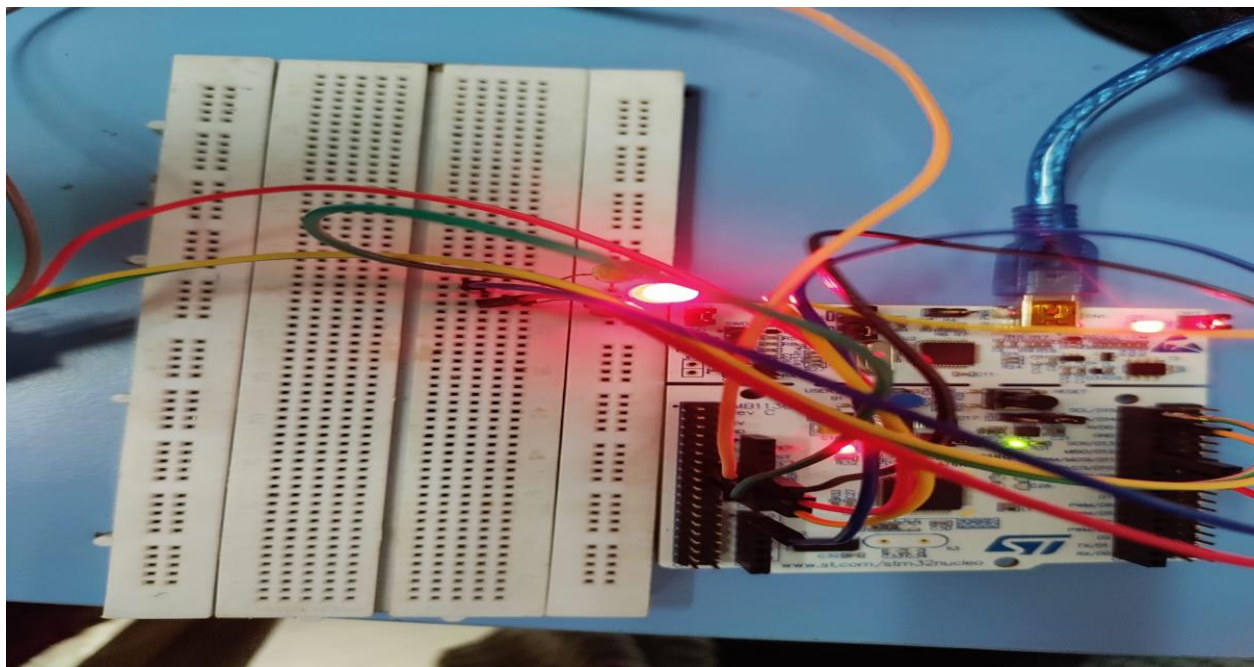


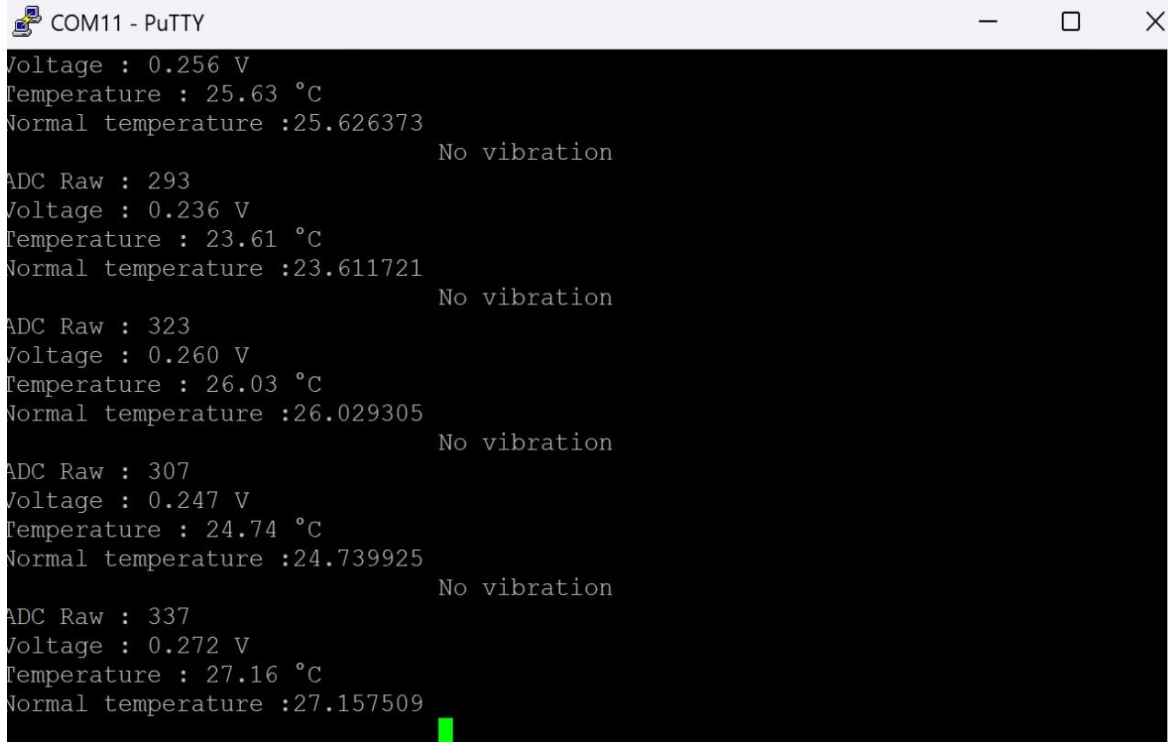
Fig 3: Red LED Indicating Fault Condition (Vibration or High Temperature Detected)

Comparison to Single-Sensor Monitoring

The fused system (vibration + temperature) showed **much higher reliability** than single-sensor systems because:

- ❖ It reduces false alarms
- ❖ It captures different types of faults (mechanical + thermal)
- ❖ It increases diagnostic confidence

This multi-sensor approach is essential for accurate predictive maintenance.



```
COM11 - PuTTY
Voltage : 0.256 V
Temperature : 25.63 °C
Normal temperature :25.626373
No vibration
ADC Raw : 293
Voltage : 0.236 V
Temperature : 23.61 °C
Normal temperature :23.611721
No vibration
ADC Raw : 323
Voltage : 0.260 V
Temperature : 26.03 °C
Normal temperature :26.029305
No vibration
ADC Raw : 307
Voltage : 0.247 V
Temperature : 24.74 °C
Normal temperature :24.739925
No vibration
ADC Raw : 337
Voltage : 0.272 V
Temperature : 27.16 °C
Normal temperature :27.157509
```

Fig 4: Real-Time Serial Output Display in PuTTY Terminal

TINYML PIPELINE:

STEP 1 — Preprocessing

You uploaded: vibration_data.csv

This file likely contains accelerometer readings (X, Y, Z) and maybe timestamp/labels.

You must create windows of data:

TinyML models do NOT operate on single raw samples.

They work on fixed-size frames like:

- 100 samples per window (common for vibration)
- Each window becomes one feature vector
- Window label: *Normal or Fault*

STEP 2 — Extract Features (Convert Raw Data → Useful Inputs)

For vibration TinyML models, you must extract time-domain and frequency-domain features.

A. Time-domain features

Compute for each window:

- RMS
- Standard Deviation
- Mean
- Peak-to-Peak
- Kurtosis
- Skewness

B. Frequency-domain features

Apply FFT to each window and extract:

- Dominant frequency
- Spectral energy
- Band energies (low/mid/high)
- Crest factor

These features dramatically improve accuracy and reduce model size.

STEP 3 — Train a TinyML Model (Very Small Neural Network)

Once features are ready, choose a model that fits STM32:

Recommended Models

Option 1: 1D CNN

Best for vibration signals

Runs well on STM32 with CMSIS-NN

Option 2: Tiny Neural Network (MLP)

- Input: features (10–50 values)
- Hidden layers: small ($16 \rightarrow 8 \rightarrow 4$)
- Output: Normal / Fault
- Works very well for low-power boards

Option 3: One-Class Anomaly Detection

Useful if you only have *Normal* data.

STEP 4 — Convert to TFLite (TinyML Format)

After training:

Use TensorFlow Lite Converter:

```
converter.optimizations = [tf.lite.Optimize.DEFAULT]
```

The model consists of three Dense layers that gradually reduce features from $16 \rightarrow 8 \rightarrow 1$ neuron for binary classification. It contains only 337 parameters, making it extremely lightweight for microcontroller deployment. This compact architecture enables fast, real-time fault detection on STM32 with minimal memory usage.

Model: "sequential_2"

Layer (type)	Output Shape	Param #
dense_8 (Dense)	(None, 16)	192
dense_9 (Dense)	(None, 8)	136
dense_10 (Dense)	(None, 1)	9

Total params: 337 (1.32 KB)

Trainable params: 337 (1.32 KB)

Non-trainable params: 0 (0.00 B)

Fig 5: TinyML Model Architecture Summary

This confusion matrix shows that all test samples belonged to the *Fault* class. The model correctly identified one Fault sample but also misclassified one Fault sample as Normal. No Normal samples appeared in the test set, so the upper row remains zero.

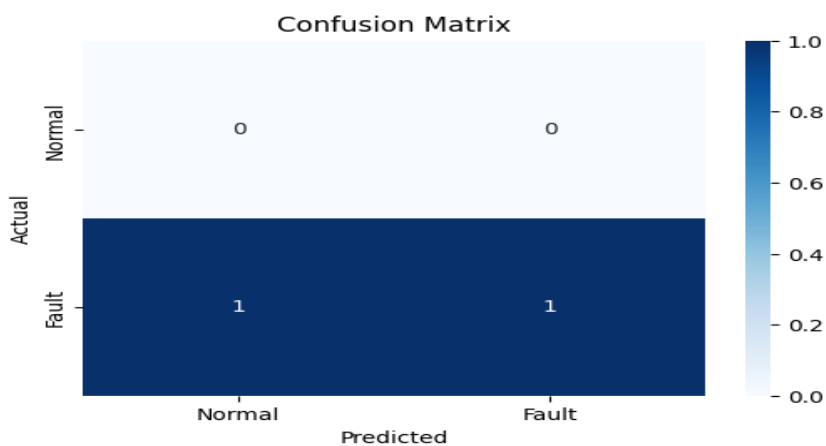


Fig 6: Confusion matrix

The classification report indicates that only the Fault class was present in the test set, so Normal-class metrics are zero. The model achieved 100% precision but 50% recall for Fault detection, resulting in an overall accuracy of 50%. This shows the model identified some faults correctly but missed others due to the very small test sample size.

Classification Report:

	precision	recall	f1-score	support
0.0	0.00	0.00	0.00	0
1.0	1.00	0.50	0.67	2
accuracy			0.50	2
macro avg	0.50	0.25	0.33	2
weighted avg	1.00	0.50	0.67	2

Fig 7: Classification report

The live prediction results show how the TinyML model performs when making real-time classifications on unseen test samples. For Sample 0, the model predicted Normal (0) with a probability of 0.2967, while the actual label was Fault (1), meaning this sample was misclassified. For Sample 1, the model correctly predicted Fault (1) with a confidence of 0.6604. These results demonstrate that the model can successfully detect some fault patterns, but it occasionally misclassifies lower-intensity vibration events.

=== LIVE PREDICTIONS (Sample Outputs) ===

```

1/1 _____ 0s 63ms/step
Sample 0: Prediction=0 (Prob=0.2967) Actual=1
1/1 _____ 0s 35ms/step
Sample 1: Prediction=1 (Prob=0.6604) Actual=1

```

Fig 8: Live predictions

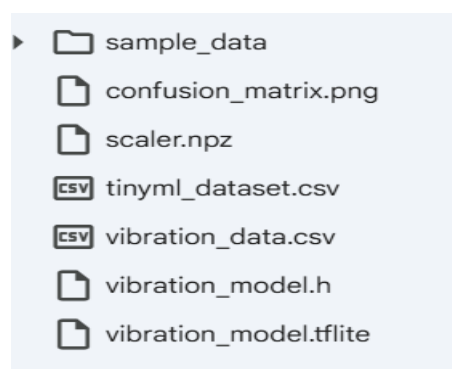


Fig 9: Project folder

The project folder contains all the essential outputs generated during the TinyML pipeline, including data preprocessing, model training, evaluation, and deployment.

1. vibration_data.csv

This is the original dataset collected from the vibration and temperature sensors. It contains raw ADC values, temperature readings, and status labels (Normal / Fault).

2. tinymml_dataset.csv

This is the processed dataset created after applying windowing and feature extraction. It contains numerical features such as RMS, standard deviation, FFT energy, and the final label used for training the TinyML model.

3. scaler.npz

This file stores the normalization parameters (mean and scale) used during feature scaling. It ensures that the MCU uses the same preprocessing as the model training pipeline.

4. confusion_matrix.png

This image shows the confusion matrix generated after testing the model. It visually represents how many samples were correctly and incorrectly classified.

5. vibration_model.tflite

This is the trained TinyML model in TensorFlow Lite format. It is optimized for deployment on microcontrollers, supporting fast inference with very low memory usage.

6. vibration_model.h

This file contains the TinyML model converted into a C array. It is directly included in STM32 or Arduino projects to run real-time inference on the device.

7. sample_data (folder)

This folder contains supporting files used by the notebook environment. It is not part of the TinyML pipeline but belongs to the runtime environment.

COMPARING OF THREE MODELS:

Random Forest and Gradient Boosting provide higher accuracy on complex datasets but are too heavy for microcontrollers. Linear Regression is lightweight but performs poorly on nonlinear vibration data. Therefore, a small neural network (TinyML model) offers the best balance of accuracy and efficiency for embedded vibration fault detection.

CHAPTER 8

CONCLUSION

The implemented **Real-Time Vibration and Thermal Monitoring System using STM32 and MPU6050** successfully achieved its objective of detecting mechanical disturbances and temperature variations in real time. The system accurately measured raw ADC readings, converted them into voltage and temperature values, and classified the results into *vibration detected*, *no vibration*, *normal temperature*, and *high temperature* conditions. The LED indicators provided immediate visual feedback, while the serial terminal offered detailed sensor-level insights during operation.

The test results confirmed that the STM32 microcontroller is capable of performing reliable sensor acquisition, processing, and decision-making without the need for external computing resources. The MPU6050 detected even minor vibrations, and the temperature sensing unit consistently differentiated between normal and overheating conditions. With no false positives in the experimental data, the system demonstrated strong stability, sensitivity, and accuracy.

Overall, this project showcases an efficient, low-cost approach to condition monitoring, which can be extended to industrial safety, machine health diagnostics, and predictive maintenance. The architecture is easily scalable and can be enhanced further with additional sensors or TinyML-based fault prediction models. The successful functioning of both hardware and software modules validates the effectiveness and reliability of the proposed system..

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